

Name: _____

Solutions

These questions assess your understanding of the material from Chapters 1, 2, 3 of OpenStax College Physics. Please write your name on the sheets of paper that you use and show all your work.

1. If a marathon runner averages 10 mi/h, how long does it take him to run a 21.6-mi marathon?

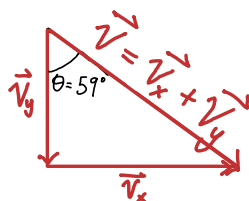
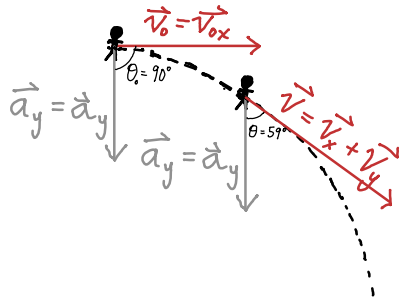
ANSWER: 2.16 hours

$$v_{avg} = \frac{\Delta x}{\Delta t} = \frac{\Delta x}{t} \quad (\because \Delta t = t)$$

$$t = \frac{\Delta x}{v_{avg}}$$

$$t = \frac{21.6 \text{ mi}}{(10 \frac{\text{mi}}{\text{h}})} = \underline{2.16 \text{ h}}$$

2. Brian, a fireman by profession but physicist at heart, jumps horizontally from a burning building with an initial speed of 9.0 m/s. Later, Brian is interested in determining the time at which the angle between his velocity and acceleration vectors is 59° . At what time does that happen?

ANSWER: 0.552 s

$$\tan \theta = \frac{|\vec{v}_y|}{|\vec{v}_x|}$$

$$\theta(t) = \tan^{-1} \left(\frac{v_y(t)}{v_x(t)} \right)$$

find t when $\theta(t) = 59^\circ$

$$v_x(t) = v_{0x} \because a_x = 0$$

$$v_y(t) = v_{0y} + a_y t$$

$$v_y(t) = 0 \frac{\text{m}}{\text{s}} + (-g)t$$

$$v_y(t) = -gt$$

$$\therefore \theta(t) = \tan^{-1} \left(\frac{v_y(t)}{v_x(t)} \right) = \tan^{-1} \left(\frac{v_{0y}}{g t} \right)$$

find t when $\theta(t) = 59^\circ$

$$\theta(t) = \tan^{-1} \left(\frac{v_{0x}}{g t} \right)$$

$$59^\circ = \tan^{-1} \left(\frac{9 \frac{\text{m}}{\text{s}}}{(9.8 \frac{\text{m}}{\text{s}^2}) t} \right)$$

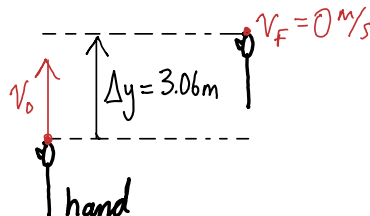
$$\tan(59^\circ) = \tan \left(\tan^{-1} \left(\frac{9 \frac{\text{m}}{\text{s}}}{(9.8 \frac{\text{m}}{\text{s}^2}) t} \right) \right)$$

$$\tan(59^\circ) = \frac{9 \frac{\text{m}}{\text{s}}}{(9.8 \frac{\text{m}}{\text{s}^2}) t}$$

$$t = \frac{9 \frac{\text{m}}{\text{s}}}{(9.8 \frac{\text{m}}{\text{s}^2}) \tan 59^\circ}$$

$$t = \underline{0.552 \text{ s}}$$

3. A basketball referee tosses the ball straight up for the starting tip-off. What is the minimum velocity at which a basketball player must leave the ground in order to rise 3.06 m above the floor and make contact with the ball?

ANSWER: 7.744 m/s

See that the velocity in the y direction is zero at the peak. This is always true.

$v_y = 0 \frac{\text{m}}{\text{s}}$ at the peak.

$$\text{i.e. } v_y(\Delta y = 3.06 \text{ m}) = 0 \frac{\text{m}}{\text{s}}$$

$$(v_y(\Delta y))^2 = v_{0y}^2 + 2a_y \Delta y$$

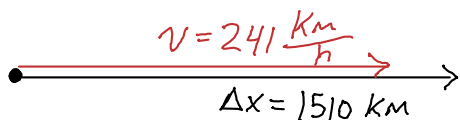
$$(0 \frac{\text{m}}{\text{s}})^2 = v_{0y}^2 + 2(-9.8 \frac{\text{m}}{\text{s}^2})(3.06 \text{ m})$$

$$v_{0y} = \pm 7.744 \frac{\text{m}}{\text{s}}$$

$$v_{0y} = 7.744 \frac{\text{m}}{\text{s}} \because \text{the basketball player jumps upward.}$$

4. A small airplane flies in a straight line at a speed of 241 km/h. How long does it take the plane to fly 1510 km?

ANSWER: 375.9 min



$$|\vec{v}| = \frac{|\Delta \vec{x}|}{\Delta t}$$

$$\Delta t = \frac{|\Delta \vec{x}|}{|\vec{v}|}$$

$$\Delta t = \frac{1510 \text{ km}}{241 \frac{\text{km}}{\text{h}}}$$

$$\Delta t = 6.266 \text{ h}$$

$$\text{or } 6.266 \text{ h} \left| \frac{60 \text{ min}}{1 \text{ h}} \right| = \underline{375.9 \text{ min}}$$

5. Chloe boards a light-rail commuter train each day for work. One day, Chloe decides to use her phone's stopwatch to measure the time it takes for the train to reach its top speed of 105.9 km/h from rest, and finds that it takes 28.9 s. What is the train's average acceleration?

ANSWER: 1.018 m/s²

$$v_f = 105.9 \frac{\text{km}}{\text{h}} \left| \frac{1000 \text{ m}}{1 \text{ km}} \right| \left| \frac{1 \text{ h}}{60 \text{ min}} \right| \left| \frac{1 \text{ min}}{60 \text{ s}} \right| = 29.42 \frac{\text{m}}{\text{s}}$$

$$|\vec{a}| = \frac{|\Delta \vec{v}|}{\Delta t}$$

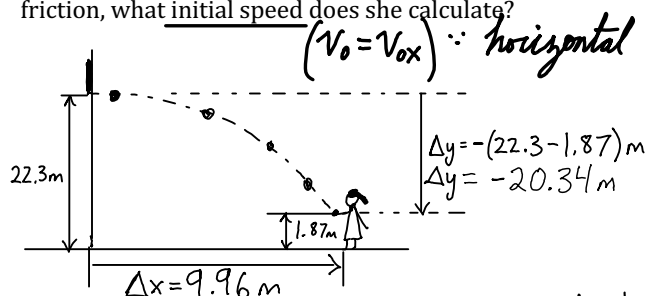
$$a_{\text{avg}} = \frac{v_f - v_0}{\Delta t}$$

$$a_{\text{avg}} = \frac{29.42 \frac{\text{m}}{\text{s}} - 0 \frac{\text{m}}{\text{s}}}{28.9 \text{ s}}$$

$$\underline{a_{\text{avg}} = 1.018 \frac{\text{m}}{\text{s}^2}}$$

6. Samantha throws a cell phone horizontally from her dorm window 22.3 m above the ground. Sarah, another student, stands 9.96 m away from the dorm and catches the cell phone at a height of 1.87 m above the ground. Sarah claims that Samantha threw the cell phone too slowly, so Samantha wants to calculate its initial speed to prove her wrong. Ignoring friction, what initial speed does she calculate?

ANSWER: 4.89 m/s



(note: $v_x(t) = v_{0x} \therefore a_x = 0$) $\therefore v_x(t) = \frac{\Delta x}{t}$

$$|\vec{v}_x| = \frac{|\Delta \vec{x}|}{\Delta t}$$

$$t = \frac{\Delta x}{v_{0x}} \quad \text{Equate these \& solve for } v_{0x}$$

$$\frac{\Delta x}{v_{0x}} = \sqrt{\frac{-2\Delta y}{g}}$$

$$v_{0x} = \Delta x \sqrt{\frac{-g}{2\Delta y}}$$

$$v_0 = v_{0x} = 9.96 \text{ m} \sqrt{\frac{-9.8 \frac{\text{m}}{\text{s}^2}}{2(-20.34 \text{ m})}} = \underline{4.89 \frac{\text{m}}{\text{s}}}$$

$$v_{fy} = v_{0y} + a_y t$$

$$\therefore v_{fy} = v_y(t) = -gt \quad (\because v_{0y} = 0 \text{ \& } a_y = -g)$$

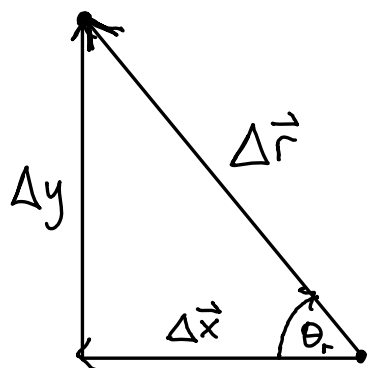
$$v_{fy}^2 = v_{0y}^2 + 2a\Delta y$$

$$v_{fy} = -\sqrt{2(-g)\Delta y}$$

$$-gt = -\sqrt{2g\Delta y}$$

$$t = \sqrt{\frac{-2\Delta y}{g}}$$

7. Out of curiosity, you want to conduct some geographic measurements of your house and your friend Aaron's. To get to Aaron's, you have to walk 2.94 km west and then 3.36 km north. How far does he live from you if you drew a straight line on the map?



find $|\Delta \vec{r}|$ ANSWER: 4.5 km

$$|\vec{r}| = \sqrt{|\Delta \vec{x}|^2 + |\Delta \vec{y}|^2}$$

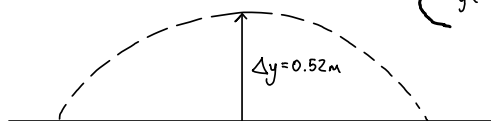
$$|\vec{r}| = \sqrt{(-2.94 \text{ km})^2 + (3.36 \text{ km})^2} = \underline{4.5 \text{ km}}$$

note: to find θ_r :

$$\theta_r = \tan^{-1}\left(\left|\frac{\Delta y}{\Delta x}\right|\right)$$

$$\theta_r = \tan^{-1}\left(\frac{3.36}{2.94}\right) = 48.8^\circ \text{ North of west (not "Northwest", which is ambiguous)}$$

8. A cricket just barely jumps over a 0.52-m-tall log. How long is the cricket in the air, measured from when it jumps to when it lands?



$(v_y(t=t_{\text{peak}}) = v_y(\Delta y = 0.52 \text{ m}) = 0)$ ANSWER: 0.652 s

$$(v(\Delta y))^2 = v_{oy}^2 + 2a_y \Delta y$$

$$0^2 = v_{oy}^2 + 2(-g)h_{\text{peak}}$$

$$v_{oy} = \sqrt{2gh_{\text{peak}}}$$

$$v_{fy} = v_{oy} + a_y t$$

$$0 = \sqrt{2gh_{\text{peak}}} - g t_{\text{peak}}$$

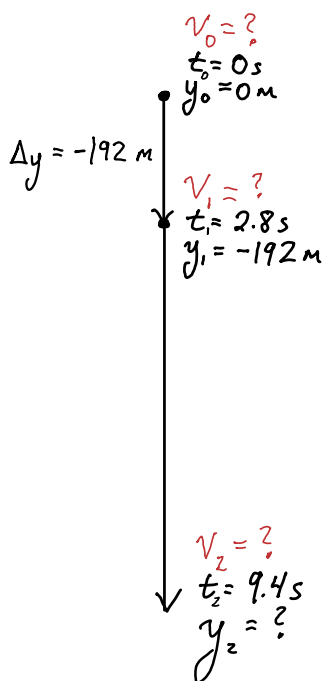
$$t_{\text{peak}} = \sqrt{\frac{2h_{\text{peak}}}{g}}$$

$$t_{\text{total}} = 2t_{\text{peak}}$$

$$t_{\text{total}} = \sqrt{8 \frac{h_{\text{peak}}}{g}} = \sqrt{8 \left(\frac{0.52 \text{ m}}{9.8 \frac{\text{m}}{\text{s}^2}}\right)} = \underline{0.652 \text{ s}}$$

9. Kevin is walking on a bridge between two mountain peaks, when curiosity suddenly overcomes him and he throws a frisbee straight down from the bridge to the distant vale below. If the frisbee falls 192 m in 2.8 s, then how far will it have fallen in 9.4 s?

ANSWER: 948.6 m



$$y(t) = y_0 + v_0 t + \frac{1}{2} a_y t^2$$

$$y_1 = y_0 + v_0 t_1 + \frac{1}{2} g t_1^2$$

$$v_0 = \frac{1}{t_1} (y_1 - y_0 + \frac{1}{2} g t_1^2) \quad (\text{find initial velocity})$$

$$y_2 = y_0 + v_0 t_2 + \frac{1}{2} g t_2^2$$

$$y_2 = y_0 + \left[\frac{1}{t_1} (y_1 - y_0 + \frac{1}{2} g t_1^2) \right] t_2 + \frac{1}{2} g t_2^2$$

$$y_2 = \frac{t_2}{t_1} \left(y_1 + \frac{1}{2} g t_1^2 \right) - \frac{1}{2} g t_2^2 \quad \because y_0 = 0 \text{ m}$$

$$y_2 = \frac{9.4 \text{ s}}{2.8 \text{ s}} \left(-192 \text{ m} + \frac{1}{2} (9.8 \frac{\text{m}}{\text{s}^2}) (2.8 \text{ s})^2 \right) - \frac{1}{2} (9.8 \frac{\text{m}}{\text{s}^2}) (9.4 \text{ s})^2$$

$$y_2 = -948.6 \text{ m}$$

this is the y position.

distance traveled is always positive.

$$|\Delta y| = \underline{948.6 \text{ m}}$$

10. Efe is sitting in his folding chair on the beach, watching his friend Ethan as he rides in a catamaran moving at a velocity of 0.84 m/s at an angle of 29.4° south of east. Ethan then passes his sister, Mary, who is riding in a barge, which, from Ethan's perspective, is travelling at a velocity of 9.98 m/s at an angle of 13.6° west of north. From Efe's perspective, what is the velocity of Mary on the barge?

Efe is the "objective" stationary observer
 $V_{E,M}$: velocity of Ethan relative to Mary
 $V_{M,E}$: Velocity of Mary relative to Ethan
 V_M' : "Objective" Velocity of Mary (relative to Efe)

Review of relative velocity:

$$\begin{array}{c} A \xrightarrow{50 \text{ mph} = V_A'} \\ B \xrightarrow{70 \text{ mph} = V_B'} \end{array} \quad \vec{V}_{B,A} = \vec{V}_B' - \vec{V}_A' \quad \text{Velocity of B relative to A}$$

ANSWER: $9.427 \frac{\text{m}}{\text{s}}, 9.9^\circ \text{ W of N}$

$$|\vec{V}_M'| = \sqrt{|\vec{V}_{Mx}'|^2 + |\vec{V}_{My}'|^2}$$

$$|\vec{V}_M'| = \sqrt{(1.6149 \frac{\text{m}}{\text{s}})^2 + (9.2878 \frac{\text{m}}{\text{s}})^2}$$

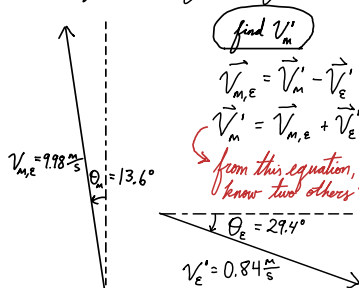
$$|\vec{V}_M'| = 9.427 \frac{\text{m}}{\text{s}}$$

$$\theta_M' = \tan^{-1} \left(\frac{|\vec{V}_{Mx}'|}{|\vec{V}_{My}'|} \right)$$

$$\theta_M' = \tan^{-1} \left(\frac{1.6149 \frac{\text{m}}{\text{s}}}{9.2878 \frac{\text{m}}{\text{s}}} \right)$$

$$\theta_M' = 9.9^\circ \text{ West of North}$$

$$\therefore \vec{V}_M' = 9.427 \frac{\text{m}}{\text{s}}, 9.9^\circ \text{ W of N}$$



find V_M'

$$\vec{V}_{M,E} = \vec{V}_M' - \vec{V}_E'$$

$$\vec{V}_M' = \vec{V}_{M,E} + \vec{V}_E'$$

from this equation, we know two others:

$$\vec{V}_{Mx}' = \vec{V}_{M,E,x} + \vec{V}_{E,x}'$$

$$\vec{V}_{Mx}' = -V_{M,E} \sin \theta_M + V_E' \cos \theta_E$$

$$\vec{V}_{Mx}' = -(9.98 \frac{\text{m}}{\text{s}}) \sin 13.6^\circ + (0.84 \frac{\text{m}}{\text{s}}) \cos 29.4^\circ$$

$$\vec{V}_{Mx}' = -1.6149 \frac{\text{m}}{\text{s}}$$

$$\vec{V}_{My}' = \vec{V}_{M,E,y} + \vec{V}_{E,y}'$$

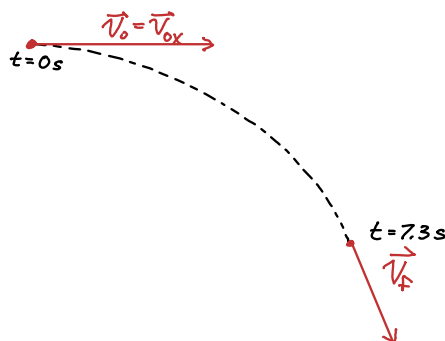
$$\vec{V}_{My}' = V_{M,E} \cos \theta_M - V_E' \sin \theta_E$$

$$\vec{V}_{My}' = 9.98 \frac{\text{m}}{\text{s}} \cos 13.6^\circ - 0.84 \sin 29.4^\circ$$

$$\vec{V}_{My}' = 9.2878 \frac{\text{m}}{\text{s}}$$

11. Natalie throws a book horizontally from a bridge with an initial speed of 20.1 m/s . Find the book's speed when it enters the water 7.3 seconds later. You may ignore air resistance.

ANSWER: $74.3 \frac{\text{m}}{\text{s}}$



$$\vec{V}_{fx} = \vec{V}_{0x} + \vec{a}_x t$$

$$\vec{V}_{fx}(t=7.3s) = 20.1 \frac{\text{m}}{\text{s}} + (0 \frac{\text{m}}{\text{s}^2})(7.3s)$$

$$\vec{V}_{fx} = 20.1 \frac{\text{m}}{\text{s}}$$

$$\vec{V}_{fy} = \vec{V}_{0y} + \vec{a}_y t$$

$$\vec{V}_{fy}(t=7.3s) = 0 \frac{\text{m}}{\text{s}} + (-9.8 \frac{\text{m}}{\text{s}^2})(7.3s)$$

$$\vec{V}_{fy} = -71.54 \frac{\text{m}}{\text{s}}$$

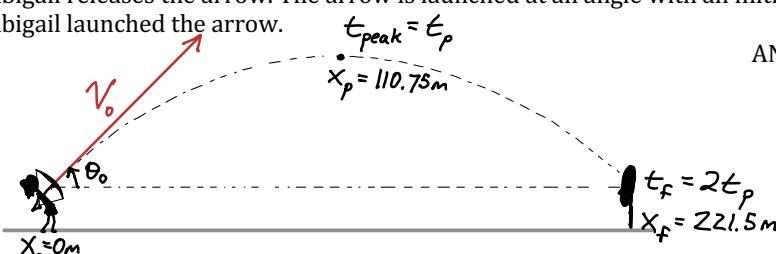
$$|\vec{V}_f| = \sqrt{V_{fx}^2 + V_{fy}^2}$$

$$|\vec{V}_f| = \sqrt{(20.1 \frac{\text{m}}{\text{s}})^2 + (71.54 \frac{\text{m}}{\text{s}})^2}$$

$$|\vec{V}_f| = 74.3 \frac{\text{m}}{\text{s}}$$

12. Abigail aims her arrow towards a target located 221.5 m away. The target's bull's eye aligns with the height from which Abigail releases the arrow. The arrow is launched at an angle with an initial speed of 47.4 m/s . Find the angle at which Abigail launched the arrow.

ANSWER: 37.5° above the horizontal



$$\sin \theta_0 \cos \theta_0 = \frac{g t_p}{V_0} \frac{X_p}{V_0 t_p}$$

See next page for work $\rightarrow \frac{1}{2} \sin 2\theta_0 = \frac{g X_p}{V_0^2}$

$$\theta_0 = \frac{1}{2} \sin^{-1} \left(\frac{2g X_p}{V_0^2} \right)$$

$$\theta_0 = \frac{1}{2} \sin^{-1} \left(\frac{2(9.8 \frac{\text{m}}{\text{s}^2})(110.75 \text{ m})}{(47.4 \frac{\text{m}}{\text{s}})^2} \right)$$

$$\theta_0 = 37.5^\circ \text{ above horizontal}$$

$$V_x(t) = V_{0x} = V_0 \cos \theta_0 \because a_x = 0$$

$$V_y(t) = V_{0y} + a_y t$$

$$V_y(t) = V_0 \sin \theta_0 - g t$$

$$V_y(t_p) = 0 = V_0 \sin \theta_0 - g t_p$$

$$\therefore t_p = \frac{V_0 \sin \theta_0}{g}$$

$$\sin \theta_0 = \frac{g t_p}{V_0}$$

$$X(t) = X_0 + V_{0x} t + \frac{1}{2} a_x t^2$$

$$X(t_p) = X_p = 0 \text{ m} + V_0 \cos \theta_0 t_p + \frac{1}{2} (0 \frac{\text{m}}{\text{s}^2}) t_p^2$$

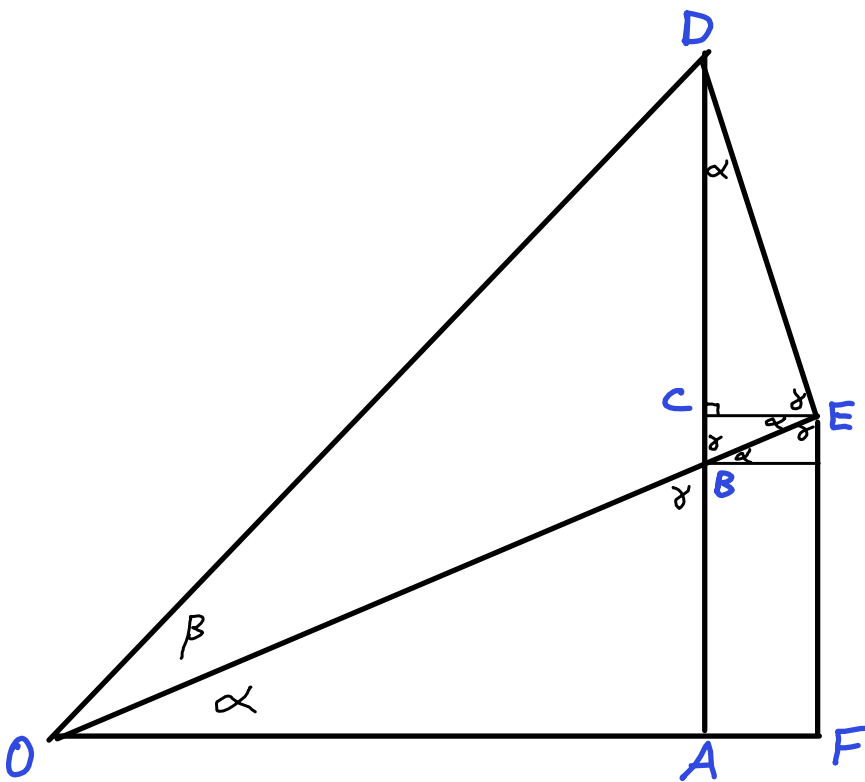
$$X_p = V_0 \cos \theta_0 t_p$$

$$t_p = \frac{X_p}{V_0 \cos \theta_0}$$

$$\cos \theta_0 = \frac{X_p}{V_0 t_p}$$

you could at this point, equate these two & solve using a trig identity

What is $\sin \theta \cos \theta$?



say $\alpha = \theta$ & $\beta = \theta$

$$\sin \theta \cos \theta = \sin \alpha \cos \beta$$

$$\sin \theta \cos \theta = \frac{EF}{EO} \frac{EO}{DO}$$

$$\sin \theta \cos \theta = \frac{EF}{DO}$$

$$\sin \theta \cos \theta = \frac{AC}{DO}$$

$$\sin \theta \cos \theta = \frac{\frac{1}{2} AD}{DO} = \frac{1}{2} \frac{AD}{DO}$$

$$\sin \theta \cos \theta = \frac{1}{2} \sin(\alpha + \beta)$$

$$\sin \theta \cos \theta = \frac{1}{2} \sin(2\theta)$$

This is just the derivation of the sine double-angle identity, which you can memorize